

M. Thufail Alwannabil Samas

Data and AI Professional | Machine Learning, Data Systems and Applied AI

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SUMMARY

Data and AI professional with experience building Python/PostgreSQL data pipelines, machine-learning and time-series systems, and controlled AI automation workflows. Built solutions for optical sensor monitoring, retail sales forecasting, local LLM-assisted service automation, and weather forecasting, spanning model evaluation, FastAPI services, Docker delivery, workflow orchestration, and human-in-the-loop controls. BNSP-certified Data Scientist and first author of a peer-reviewed XGBoost rainfall forecasting publication.

EXPERIENCE

PT Applikanusa Lintasarta

Surabaya, Indonesia

Junior Programmer

January 2026 - April 2026

- Developed a project-stage DWDM optical monitoring and assurance system for approximately 400 directional sensors, integrating five-minute PRTG telemetry with ENIMS network metadata to identify operational failures, persistent signal shifts, gradual degradation, and investigation priorities.
- Built an incremental Python and PostgreSQL pipeline to synchronize metadata, collect telemetry, repair data gaps, generate processed observations, detect events, compare paired sensors, rank sensor priorities, and serve dashboard data through an API.
- Implemented data-quality and failure-detection rules for missing readings, low optical power, downtime, and zero coverage, grouping adjacent five-minute failure intervals into continuous events and excluding invalid observations from statistical analysis.
- Developed sensor-adaptive analytics using robust six-hour medians, PELT change-point detection, Theil-Sen slope estimation, and Mann-Kendall trend testing to distinguish persistent signal shifts and gradual degradation from short-term telemetry noise.
- Created a browser-based monitoring interface with Operations Board, Event Explorer, and Management Overview views, allowing users to review active failures, statistical events, paired-sensor evidence, investigation charts, and current sensor priorities.

SMP GIKI 3 Surabaya

Surabaya, Indonesia

Coding and AI Teacher

August 2025 - December 2025

- Taught Python fundamentals to Grade 8 and 9 students, covering variables, data types, operators, conditionals, loops, and programming logic through structured lessons and hands-on coding exercises.
- Introduced foundational AI concepts using Teachable Machine, guiding students through datasets, model training, and image-classification experiments to demonstrate how machine-learning models work in practice.
- Developed lesson materials, coding exercises, quizzes, and sample programs, using practical assignments and structured assessments to support instruction and evaluate student understanding.

Universitas Islam Negeri Sunan Ampel Surabaya (UINSA)

Surabaya, Indonesia

Programming Teaching Assistant

August 2022 - July 2024

- Guided undergraduate students across four semesters of programming courses in MATLAB, Python, and computational problem-solving through practical exercises and assignment support.
- Supported students in debugging, data handling, and algorithmic thinking, helping them turn mathematical and computational concepts into working code for exercises and assignments.
- Developed introductory machine-learning materials covering Linear Regression, K-Nearest Neighbors, and K-Means clustering, and provided structured technical feedback on student assignments.

Stasiun Meteorologi Maritim Kelas II Tanjung Perak Surabaya (BMKG)

Surabaya, Indonesia

Data Science Intern

February 2024 - June 2024

- Developed a one-page extreme-weather early-warning dashboard using HTML, CSS, JavaScript, PHP, MySQL, and periodic API updates for BMKG's forecasting workflow.
- Processed and compared BMKG cloud-prediction imagery with Himawari IR satellite imagery using Python, supporting descriptive assessment of cloud movement and temperature patterns.
- Built a three-layer BiLSTM model for next-day rainfall forecasting using seven-day multivariate sequences, a focused grid search, and an 80:20 chronological split, achieving MAAPE of 0.81 and RMSE of 10.27 mm.

EDUCATION

Universitas Islam Negeri Sunan Ampel Surabaya (UINSA)

Surabaya, Indonesia

Bachelor of Mathematics (S.Mat.)

August 2021 - January 2025

- **GPA:** 3.46/4.00
- **Undergraduate Thesis:** Rainfall Forecasting in Sumenep Regency Using XGBoost and Grid Search.
- **Academic Highlight:** Completed the 4-year bachelor's program in 7 semesters (3.5 years).
- **Relevant Coursework:** Artificial Intelligence, Programming, Mathematical Statistics, Statistical Methods, Multivariate Analysis, Numerical Methods, and Applied Linear Algebra.

SKILLS

- **Programming & Query:** Python, SQL, R, MATLAB
- **Data Processing & Databases:** Pandas, NumPy, PostgreSQL, MySQL, pgvector, psycopg, Data Ingestion, Data Validation, Data Transformation, Incremental Data Pipelines
- **Machine Learning & Statistics:** scikit-learn, XGBoost, TensorFlow/Keras, statsmodels, Feature Engineering, Statistical Analysis, Model Evaluation, Cross-Validation, Time-Series Forecasting
- **Applied AI & Automation:** FastAPI, n8n, Ollama, RAG, Embeddings, Vector Search, Structured LLM Output, REST APIs, Human-in-the-Loop Workflows, Idempotency, Retry and Recovery Workflows
- **Development & Visualization:** Docker, Docker Compose, Git, GitHub, GitHub Actions, CI/CD, unittest, Jupyter Notebook, Model Serving, Batch Inference, HTML, CSS, JavaScript

PROJECTS

AI Service Request Automation

Addressed manual and inconsistent handling of employee IT service requests by building an end-to-end local AI automation system using FastAPI, n8n, PostgreSQL/pgvector, Ollama, and Docker Compose for bilingual request classification, catalog matching, grounded policy retrieval, deterministic validation, human approval, and reliable retry/recovery workflows. The system passed a frozen 40-case acceptance gate with 100.0% classification macro F1, required-field accuracy, retrieval Recall@3, and citation validity, while achieving 96.7% route and semantic task success and enabling 90.0% of semantic requests to proceed without service-agent triage review.

Retail Sales Forecasting & Planning System

Addressed the challenge of forecasting daily sales across 54 stores and 33 product families by developing an end-to-end 16-day forecasting pipeline on more than 3 million labeled rows, using chronological validation to compare Ridge and XGBoost models. Selected XGBoost with an internal-test WAPE of 15.6431%, packaged the model as a versioned artifact for batch and FastAPI inference, generated 28,512 forecasts, and validated consistent predictions through 44/44 automated contract tests in GitHub Actions across notebook, batch, API, and Docker execution.

Thyroid Cancer Recurrence Classification

Addressed the problem of identifying patients at risk of thyroid cancer recurrence by developing a reproducible XGBoost classification pipeline on 383 UCI records, incorporating mixed-type preprocessing, fold-local Information Gain feature selection, hyperparameter tuning with Grid Search, and 10-fold cross-validation. The final model achieved 97.25% pooled out-of-fold accuracy, 95.37% F1-score, 98.05% specificity, and 98.60% ROC-AUC.

PUBLICATIONS

Prediksi Curah Hujan di Kabupaten Sumenep Menggunakan Metode *Extreme Gradient Boosting* (XGBoost) dan Algoritma *Grid Search*

First Author | MIND Journal, Vol. 11, No. 1, pp. 30-43 | June 2026 | DOI: 10.26760/mindjournal.v11i1.30-43

CERTIFICATIONS

Data Scientist - Badan Nasional Sertifikasi Profesi (BNSP) | September 2024 - September 2027